



SardinianVoxes: A Speech Recognition Dataset for the Sardinian Languages

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Abstract

Low-resource languages inherently lack the extensive data typically required to train effective intelligent systems, leading to a substantial gap in technological support. In this paper, we address this challenge by focusing on the language varieties spoken in Sardinia, which are marked by limited and fragmented linguistic resources. First, we present the design and implementation of a reproducible pipeline that led to the preparation of *SardinianVoxes*, an audio-text dataset comprising approximately 170 hours of transcribed speech, carefully annotated to reflect the internal linguistic diversity of Sardinian. Second, we introduce a tailored evaluation protocol and benchmarking suite aimed at assessing the performance of state-of-the-art speech-to-text models, and their fine-tuned counterparts, across Sardinian varieties. We expect that our contributions will support the development of intelligent speech technologies for Sardinian. Source code: github.com/ai4limba/sardinian-voxes.

Index Terms: Low-Resource Languages; Minority Language; Speech-to-Text; Language Resource Development.

1. Introduction

Languages, particularly those spoken by minority or local communities, offer a unique lens through which to understand the world, as they embody the history, traditions, and cultural identity of their speakers. However, the ongoing loss of these languages often results in the irreversible disappearance of distinct worldviews and the knowledge they encode [1]. The preservation and valorization of such languages are essential for safeguarding linguistic diversity and protecting cultural heritage [2]. Additionally, promoting their use can contribute to social cohesion and inclusivity, while fostering recognition of cultural identities and enriching the collective human experience.

Intelligent systems can offer new opportunities for supporting the preservation and valorization of minority or local languages, through applications such as speech recognition, text-to-speech synthesis, and machine translation that can enable documentation, accessibility, and revitalization at scale. However, current advances behind these advanced applications have largely focused on high-resource languages, such as English, Chinese, or French, which benefit from abundant textual and speech data for model training and evaluation [3]. In contrast, low-resource languages¹, including many spoken in minority or local communities, face challenges due to limited, fragmented, or non-digitized linguistic resources and shrinking speaker pop-

ulations. This data imbalance limits the reach of intelligent systems and reinforces the technological marginalization of these languages. Timely reversing this trend calls for immediate efforts to develop high-quality, representative datasets, adapt existing models to low-resource scenarios, and design evaluation protocols that respect linguistic and cultural specificity [4].

One notable technology with the potential to advance the (digital) presence of low-resource languages is speech-to-text (STT). By converting spoken input into written text, STT systems can facilitate the creation of transcribed corpora, support language education tools, and enable voice-based interfaces tailored to local linguistic contexts, as examples. These capabilities are particularly valuable for languages with strong oral traditions, helping to ensure their continuity and visibility in the digital age [5, 6]. Sardinian² offers a compelling case: despite its active use across multiple varieties, its linguistic resources are sparse, fragmented, inconsistent, and limited in volume to support modern, intelligent STT models training [7] and recent studies underscore the need for Sardinian-specific tools [8].

To this end, in this paper, we address the limitations outlined above through two main novel contributions: (i) the design and implementation of a reproducible pipeline for the preparation of *SardinianVoxes*, an audio-text dataset comprising approximately 170 hours of transcribed speech, with coverage of the diverse, annotated varieties of the Sardinian language; and (ii) the design and development of a tailored evaluation protocol and benchmarking suite for assessing the performance of state-of-the-art STT models across Sardinian varieties. Our study lays the groundwork for a standardized framework to develop speech technologies for Sardinian, also beyond STT.

2. Related Work

This work builds upon two foundational areas: (i) the advancement of deep learning approaches for speech recognition, and (ii) the design and curation of speech datasets for low-resource languages. We briefly review key developments in both areas.

2.1. Deep Learning for Speech Recognition

Early approaches to STT primarily relied on statistical models, notably Hidden Markov Models (HMMs) [9]. With the advent of deep learning [10], substantial progress has been made, particularly through the use of architectures such as Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and Transformers. Additionally, traditional STT systems were typically built as multi-stage pipelines. In contrast, recent advances have shifted toward end-to-end models capable of di-

¹In this work, we use the term *low-resource languages* to refer to minority or local languages that lack sufficient digital or annotated linguistic resources for developing modern computational tools.

²In this work, we conventionally use the term *Sardinian languages* to refer broadly to minority languages spoken in Sardinia.

Table 1: Overview of publicly available multilingual speech datasets with relevance to low-resource languages. The comparison highlights their scope, properties, and known limitations, highlighting the lack of comprehensive resources for languages like Sardinian.

Dataset	Size	Properties	Strengths	Weaknesses
Common Voice	~10-100k hours	Multilingual, crowd-sourced	Open-source, diverse languages, community-driven	Inconsistent quality, limited speaker diversity for some languages
OpenSLR	Varies (small to medium)	Speech and text pairs	Free and open, covers many low-resource languages	Limited size for some languages, uneven quality
Google Speech Commands	~100 hours	Short commands, isolated words	Easy to use, well-documented	Limited to short phrases, not conversational
ELRA Catalogue	Varies (small to medium)	Multilingual, diverse formats	Covers many low-resource languages, high-quality	Expensive, limited access for some datasets
CMU Wilderness	~10-100 hours per language	Multilingual, diverse formats	Covers over 700 languages, including endangered ones	Small size for many languages, uneven quality
VoxPopuli	~10-100 hours per language	Multilingual	Covers 18 European Languages	Small size for many languages

rectly mapping speech signals to text. For instance, DeepSpeech [11] employs a tailored loss to align input and output sequences without requiring frame-level annotation. Moreover, sequence-to-sequence models, such as LAS [12], incorporate attention mechanisms that allow the model to focus on the most relevant segments to improve accuracy. An important advancement of this model class is the Transformer architecture, which employs self-attention mechanisms to model long-range dependencies in speech signals. Conformer [13] enhances this capability by combining convolutional and self-attention layers to capture both local and global context.

In low-resource language settings, where annotated speech data is scarce, models such as Wav2Vec [14] and HuBERT [15] have proven particularly valuable. These approaches learn meaningful representations from raw audio by predicting masked portions of the input signal. Once pretrained, they can be fine-tuned on smaller labeled datasets, achieving competitive performance with limited supervision. More recently, multilingual models like Whisper [16] and XLS-R [17] have been developed to support a wide array of languages and varieties. These models exploit cross-lingual transfer via shared representations, enabling zero- or few-shot learning in low-resource scenarios.

2.2. Speech Datasets for Low-Resource Languages

To better understand the landscape of available resources and the challenges specific to our study, Table 1 presents a comparative overview of existing speech datasets which also include low-languages. Despite the valuable contributions these datasets offer to the field, the datasets for many languages remain limited in scale or suffer from issues like low audio quality, sparse annotations, or a lack of linguistic diversity, hindering the development and evaluation of robust STT models. Notably, the only existing dataset for Sardinian is a small subset available through the Mozilla’s Common Voice project, containing approx. 3 hours of audio. This minimal coverage is far from sufficient to support the training of modern speech models. As such, Sardinian remains underrepresented in the speech technology landscape, highlighting the need for the dataset creation pipeline and evaluation framework proposed in this work.

3. Methodology

We emphasize that our proposal pursues two main goals. First, we present a reproducible pipeline for constructing a dataset of Sardinian speech, consisting of audio segments aligned with textual transcriptions and covering all major varieties spoken on the island. Second, we fine-tune state-of-the-art STT models on this curated dataset and evaluate their performance.

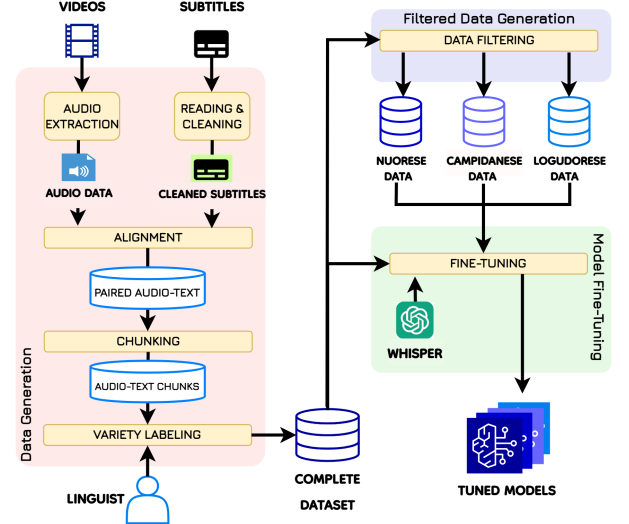


Figure 1: Our methodology for building and evaluating on the SardinianVoxes dataset. The pipeline includes (left) data generation from subtitle-aligned videos and (right) data filtering and fine-tuning of STT models on Sardinian varieties.

3.1. High-Level Pipeline Description

Figure 1 shows the steps of our pipeline, designed to meet the outlined objectives. In the first stage (Data Generation), we assumed to source and process videos with associated subtitles from a publicly accessible digital library³, to obtain aligned audio-text segments. Each segment was then expected to be manually labeled with its corresponding variety by a committee of expert linguists with recognized expertise in Sardinian⁴. The next stage (Filtered Data Generation) involved organizing the annotated segments into curated datasets for each distinct Sardinian variety. These datasets were then used to fine-tune multiple instances of a pre-trained STT model (Model Fine-Tuning). Each stage of our proposed pipeline is described in detail in the following subsections.

3.2. Part 1: Data Generation

The first part focused on the acquisition and preparation of audio-text data. To this end, we assumed to begin by collecting public videos from a digital archive dedicated to the Sardinian

³As a guiding example, it can be considered the Sardegna Digital Library: <https://www.sardegna.digitallibrary.it/en>.

⁴The annotation was carried out by a committee of academic linguists, each with over seven years of experience in Sardinian language.

language. Each video is expected to feature a person coming from a distinct geographical area of Sardinia, resulting in a collection that spans a broad spectrum of Sardinian varieties. Subtitles associated with these videos were also retrieved. The subtitle content, combined with the video titles and metadata, provided insights into each speaker’s village of origin, reinforcing the integrity of the dataset. After acquisition, the videos were processed to extract audio in .wav format, minimizing information loss. Subtitles in .vtt format were cleaned and normalized to remove inconsistencies, and their embedded timestamps were used to align them with the corresponding audio.

Given that many pre-trained models used for fine-tuning expect 30-second audio inputs, we chunked each audio-transcript pair accordingly. Specifically, using the subtitle timestamps, the speech and text were segmented into chunks of up to 30 seconds, preserving sentence integrity and ensuring that no spoken words were cut mid-utterance. Due to the high variability among Sardinian varieties, we involved again the same expert linguists to label each segment based on the variety spoken. The annotation considered the three most prominent Sardinian varieties (Campidanese, Logudorese, and Nuorese), while also including two additional labels: Other (for less common Sardinian varieties), and Non-Sardinian (e.g., other languages spoken in Sardinia, having different linguistic roots).

3.3. Part 2: Filtered Data Generation

Based on the labels provided by domain experts, we curated two distinct datasets: one encompassing all language varieties spoken in Sardinia, and another focused exclusively on the three main Sardinian varieties: Campidanese, Logudorese, and Nuorese. This was achieved by filtering the labeled data according to the assigned variety tags in the metadata file, ensuring each dataset met the intended linguistic scope.

3.4. Part 3: Model Fine-Tuning

To evaluate the effectiveness and reliability of the two generated datasets, we adopted Whisper [16], a state-of-the-art multilingual model for STT tasks. Each dataset was used to fine-tune a separate instance of the model, enabling a comparative evaluation across distinct experimental settings. This setup allowed us to assess the model’s adaptability to the level of specificity.

4. Experiments

In this section, we present an analysis of the generated datasets, describe the experimental setup adopted for the STT task, and discuss the results obtained from our benchmarking.

4.1. Example Pipeline Instantiation

We instantiated the proposed pipeline to construct a Sardinian dataset. Specifically, we leveraged publicly available material from the Sardinian Digital Library, gathering 341 video interviews of varying durations, typically between 30 and 35 minutes. Each video features a different speaker from a distinct area of Sardinia. For each resulting audio chunk, we generated corresponding metadata, including a unique identifier, the speaker’s place of origin, and spoken language variety. The final dataset comprises approximately 170 hours of transcribed speech data. Note that, to respect the terms of use of the original source, we do not redistribute the media files. Instead, we release the complete codebase required to reproduce the dataset, thereby ensuring transparency, compliance, and reproducibility.

4.2. Exploratory Data Analysis

In a first analysis, we explored the characteristics of the generated dataset to inform the fine-tuning strategy for STT models. In particular, we focused on two key aspects: (i) the length of the transcribed sentences, and (ii) the number of audio segments available for each variety. The former helps assess dataset consistency by identifying audio segments that may contain excessively long silences or insufficiently informative transcriptions. The latter provides insight into the distribution of linguistic varieties, allowing us to adapt the fine-tuning process accordingly.

Transcribed Sentence Length. Figure 2 shows the distribution of token counts per audio chunk. The data follows an approximately Gaussian shape with a slight rightward skew, indicating that most chunks contain a number of tokens close to the average, with few outliers. This observation suggests a relatively balanced dataset in terms of transcription length, supporting stable and effective training of STT models.

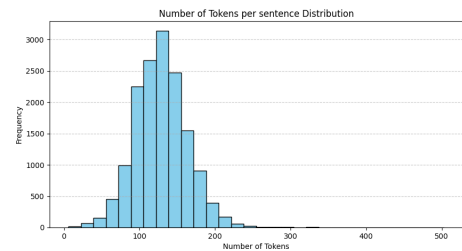


Figure 2: Distribution of the number of tokens per audio chunk.

Language Variety Distribution. Figure 3 shows the distribution of linguistic varieties. While we do not have direct control over the original data distribution, this predominance can be reasonably conjectured to stem from the higher population density in the Campidano area, where this variety is primarily spoken. This imbalance should be considered and, if possible, addressed during model fine-tuning and evaluation phases to ensure a balanced performance across varieties.

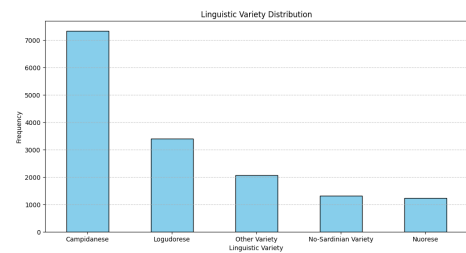


Figure 3: Distribution of linguistic varieties in the dataset.

4.3. Data Partitioning for Model Benchmarking

To establish a solid reference for future research on Sardinian, it is essential to define an appropriate train/validation/test split of the data. Given the nature of our dataset, we adopted two different, complementary splitting strategies. First, the global dataset (SardFull) was randomly divided into training, validation, and test sets, using a standard 80%-10%-10% ratio. Second, we created a filtered dataset (SardTriV) by retaining only the three most prominent Sardinian varieties (Campidanese, Logudorese, and Nuorese) and applied a stratified splitting strategy to it to ensure that the distribution of linguistic varieties is preserved across training, validation, and test sets.

4.4. Evaluation Protocol for Model Benchmarking

One of the key objectives of our work is to evaluate the performance of pre-trained STT models, specifically those that have had little or no exposure to low-resource languages, under different fine-tuning setups. To this end, we conducted an experimentation phase involving several versions of the Whisper model. We assessed performance using standard metrics in STT tasks, namely WER (Word Error Rate) [18] and the CER (Character Error Rate) [19]. The evaluation followed a three-fold strategy⁵: (i) using each model in its original, pre-trained form; (ii) fine-tuning each model on the training set of the full dataset (*SardFull*); and (iii) tuning on a subset (*SardTriV*) that includes only the three most prominent Sardinian varieties. In the third scenario, we evaluated model performance not only across the entire test set but also separately for each linguistic variety, to explore variety-specific behavior and generalization.

4.5. Benchmarking Results

In this subsection, we report the outcomes of our evaluation aimed at benchmarking the effectiveness of the fine-tuning strategies on Whisper models using our Sardinian datasets.

Fine-tuning with the Full Dataset (*SardFull*). We evaluated the performance on a more heterogeneous yet low-resource setting. Table 2 reports the results on the *SardFull* setting.

Table 2: Word and Character Error Rates (WER and CER) of Whisper models (small, medium, large) before and after fine-tuning on the *SardFull* setting (the full Sardinian dataset).

Training Set	Test Set	Model	WER	CER
-	SardFull	W-small	1.4160	0.9219
-	SardFull	W-medium	1.3543	0.8809
-	SardFull	W-large	1.1868	0.7693
SardFull	SardFull	W-small	0.6937	0.3752
SardFull	SardFull	W-medium	0.6735	0.3723
SardFull	SardFull	W-large	0.6274	0.3573

Fine-tuning clearly yields substantial improvements, reducing error metrics by approximately half and bringing performance to satisfactory levels. Notably, despite the considerable difference in model sizes, the smaller variants perform comparably to the large model. This suggests that increasing the quantity, quality, and consistency of training data can enable smaller or moderately sized models to achieve competitive results.

Fine-tuning with the Principal Varieties (*SardTriV*). Table 3 presents the performance of Whisper-based models tuned and evaluated on the *SardTriV* setting. While this setting is more linguistically homogeneous, it contains fewer samples, which limits the learning capacity. This limitation is evident in the results: although models show gains over their pre-trained versions, the reduced data volume impacts performance, particularly for larger versions. These findings suggest that larger models may be more sensitive to data scarcity, whereas smaller models tend to generalize better under such constraints.

Despite these limitations, the results are noteworthy, as they show meaningful improvements. To support this, Table 4

⁵All experiments were conducted using identical hyperparameters to ensure a fair comparison across models. Specifically, we trained for five epochs with a batch size of 4 and a learning rate of 0.0001. All runs were executed on an NVIDIA RTX A6000 GPU with 48GB of VRAM.

Table 3: Word and Character Error Rates (WER and CER) of Whisper models (small, medium, large) before and after fine-tuning on the *SardTriV* setting (three most prominent varieties).

Training Set	Test Set	Model	WER	CER
-	SardTriV	W-small	1.3730	0.9004
-	SardTriV	W-medium	1.3640	0.8968
-	SardTriV	W-large	1.1727	0.7574
SardTriV	SardTriV	W-small	0.6635	0.3471
SardTriV	SardTriV	W-medium	0.6385	0.3481
SardTriV	SardTriV	W-large	0.7430	0.4420

Table 4: Word and Character Error Rates (WER and CER) of the Whisper-medium model tuned on *SardTriV*, evaluated separately on each Sardinian variety in the *SardTriV* test partition.

Training Set	Test Set	Variety	WER	CER
-	SardTriV	Nuorese	1.3863	0.8493
-	SardTriV	Logudorese	1.3236	0.8259
-	SardTriV	Campidanese	1.3929	0.9426
SardTriV	SardTriV	Nuorese	0.6106	0.3106
SardTriV	SardTriV	Logudorese	0.631	0.3282
SardTriV	SardTriV	Campidanese	0.6463	0.3633

presents the performance of Whisper *medium* models when evaluated separately on each variety in the *SardTriV* setting.

To contextualize our findings, we discuss them with respect to those reported in [7], which evaluated multiple models on a Campidanese-only dataset (not publicly available). In their study, the best-performing Whisper *medium* model, after fine-tuning, achieved a WER of 0.79. By contrast, our fine-tuned counterpart obtained a lower WER (0.61) on the same variety (a +22.78% gain). While the comparison is not fully equivalent due to differences in datasets and setups, it nonetheless reinforces the effectiveness of our design and fine-tuning strategy in supporting speech recognition for low-resource languages.

5. Conclusion and Future Work

In this paper, we proposed *SardinianVoxes*, a speech recognition dataset aimed to address the under-representation of low-resource languages in speech recognition. The dataset captures multiple linguistic varieties spoken across Sardinia, fulfilling our first contribution: the design, implementation, and adoption of a reproducible pipeline for preparing data to train models on Sardinian speech recognition. As a second contribution, we designed a framework aimed at standardizing the model benchmarking for this language. Finally, we validated it with several versions of the Whisper model, showing that fine-tuning with Sardinian-specific data can improve performance.

Nevertheless, the findings highlight several areas requiring further investigation. First, the limited coverage across varieties suggests the need for more balanced data collection. Second, the persistence of errors in linguistically complex scenarios points to the importance of exploring advanced audio preprocessing techniques to enhance input quality. Third, the reliance on fixed chunking strategies may constrain model effectiveness, indicating that alternative segmentation methods merit evaluation. Lastly, assessing a broader range of models and training new ones from scratch could yield deeper insights into the most effective strategies for low-resource speech recognition.

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